

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

1. Rural Branch Office – Internet Connectivity and Secure Network

Problem understanding and constraints:

- High-speed wired internet is unavailable.
- Supports cloud applications and video conferencing.
- Requires secure communication with headquarters.
- Budget is limited.
- Business operations should continue with minimum interruption.
- The network should support future expansion.

Recommended solution:

For a remote rural office, I recommend a 4G/5G wireless broadband connection as the primary connection, with satellite internet as a backup if cellular coverage is unreliable.

Networking devices:

1. 5G/4G router – Provides internet connectivity through the cellular network.
2. Managed switch – Connects computers, printers, and other wired devices.
3. Wireless access point – Provides Wi-Fi for employees.
4. VPN router/firewall – Creates a secure connection with headquarters.
5. UPS – Provides temporary power during power interruptions.

Security mechanisms:

- Site-to-site IPsec VPN between the branch and headquarters.
- Firewall to block unauthorized traffic.
- WPA3 for secure Wi-Fi access.
- Strong passwords and user authentication.
- Regular firmware and security updates.

Constraint analysis:

Constraint	Solution
Limited bandwidth	Use QoS to prioritize video conferencing and business applications
Unreliable connectivity	4G/5G primary + satellite backup
Limited budget	Use cost-effective network devices and avoid unnecessary equipment
Security	Firewall + VPN + WPA3
Future scalability	Managed switch and scalable router with additional ports

Conclusion:

A 4G/5G-based network with satellite backup, firewall, VPN, managed switch, and QoS provides a good balance between cost, reliability, security, and future scalability. It allows employees to use cloud services and communicate securely with headquarters even in a remote location.

2. University Wireless Network for 3,000+ Students

Problem understanding and constraints:

- More than 3,000 students need simultaneous wireless access.
- Coverage is required in classrooms, laboratories, libraries, and hostels.
- Limited number of access points can be installed.
- Signal interference must be minimized.
- Network must provide secure authentication.
- The design must balance performance and cost.

Proposed wireless design:

The university should use centrally managed Wi-Fi 6/6E access points. Wi-Fi 6 is suitable because it supports a large number of simultaneous users efficiently.

Access point placement:

- Install APs in classrooms and laboratories based on user density.
- Place APs in corridors/common areas to cover multiple rooms where possible.
- Install additional APs in hostels and libraries because these areas have high user density.
- Avoid placing APs close to metal objects, thick concrete walls, and other sources of interference.
- Use a wireless controller/cloud management system to monitor APs.

Frequency band selection:

- 5 GHz: Primary band for high-speed connections and less interference.
- 2.4 GHz: Used mainly for devices with longer range or older compatibility.
- 6 GHz: Can be used where Wi-Fi 6E-compatible devices are available.
- Use different non-overlapping channels between nearby APs to reduce interference.

Authentication and security:

- Use WPA3-Enterprise for secure wireless access.

- Use 802.1X authentication with a RADIUS server.
- Provide separate networks/VLANs for students, faculty, guests, and administration.
- Use strong access policies and monitoring.

Constraint analysis:

Constraint	Solution
3,000+ users	Wi-Fi 6 APs with high client capacity
Limited APs	Proper AP placement based on user density
Interference	Channel planning and 5 GHz/6 GHz usage
Security	WPA3-Enterprise + 802.1X/RADIUS
Cost	Prioritize APs in high-density areas
Scalability	Centralized controller and VLAN-based design

Conclusion:

A Wi-Fi 6 centrally managed network, with careful AP placement, channel planning, 5 GHz as the primary band, and WPA3-Enterprise authentication, can provide reliable campus-wide connectivity. Instead of simply increasing the number of APs, the university can maximize performance by placing limited APs in high-density areas and managing them centrally.

3. Financial Institution – Secure Network Architecture

Problem understanding and constraints:

- Sensitive customer information must be protected.
- Employees require access to banking applications.
- Strong cybersecurity is required.
- The organization must comply with security policies and regulations.
- Security controls should not significantly reduce performance.
- The solution must remain within the operational budget.

Proposed network architecture:

The institution should use a segmented, firewall-protected network.

Network segmentation:

The network can be divided into separate VLANs/zones:

1. Employee VLAN – Normal employee computers.
2. Server VLAN – Internal business applications and databases.
3. Customer-data VLAN – Highly sensitive customer information.
4. DMZ – Public-facing online banking/web servers.
5. Guest VLAN – Internet-only access for visitors.
6. Management VLAN – Network administration devices.

This prevents an attacker who compromises one section from easily accessing sensitive systems.

Firewall deployment:

- Deploy a next-generation firewall (NGFW) at the internet boundary.
- Use firewall rules to control incoming and outgoing traffic.
- Place public-facing servers in a DMZ.
- Use internal firewall controls between critical network segments.

Authentication:

- Use multi-factor authentication (MFA) for sensitive applications.
- Use 802.1X for network access control.

- Apply role-based access control so employees can access only the information required for their jobs.
- Use strong passwords and account policies.

Intrusion detection:

- Deploy IDS/IPS to monitor suspicious network traffic.
- Generate alerts for unusual login attempts, malware, port scanning, and unauthorized access.
- Maintain security logs for auditing and compliance.

Constraint analysis:

Constraint	Solution
High security	VLANs, firewalls, IDS/IPS and MFA
Sensitive data	Separate customer-data network
Performance	Use high-performance firewall and efficient security rules
Compliance	Logging, access control and regular security audits
Budget	Use centralized security management and prioritize critical systems
Availability	Redundant firewalls and network devices where required

Simple architecture:

Internet → Edge Router → Firewall → DMZ / Core Switch → VLANs

- DMZ: Online banking/web servers
- VLAN 10: Employees
- VLAN 20: Application servers
- VLAN 30: Customer database
- VLAN 40: Guest network
- VLAN 50: Network management
- IDS/IPS: Monitors traffic across critical network segments

Conclusion:

A segmented network with VLANs, DMZ, next-generation firewall, MFA, 802.1X, and IDS/IPS provides strong protection for sensitive financial data. The design also improves compliance and limits the impact of security breaches while maintaining good network performance.